Question Paper

Exam Date & Time: 14-Apr-2025 (01:15 PM - 04:30 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Theory Examination April 2025
Sixth Semester of B.Tech.

DESIGN OF LANGUAGE PROCESSOR [CSE313]

Marks: 70 **Duration: 195 mins.**

Section 1

Answer all the questions. Section Duration: 40 mins

1 The number of tokens in the following C statement is (1)

printf("i=%d, &i=%x", i,&i);

9 10 8 11

Course Outcome: "CO1" / Cognitive Level: "Understand"

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2 Match all items in Group 1 with correct options from those given in Group 2. (1)

Group 1	Group 2
P. Regular expression	1. Syntax analysis
Q. Pushdown automata	2. Code generation
R. Dataflow analysis	3. Lexical analysis
S. Register allocation	4. Code optimization

P-4, Q-1, R-2, S-3 P-3, Q-1, R-4, S-2 P-3, Q-4, R-1, S-2 P-2, Q-1, R-4, S-3

Course Outcome: "CO1" / Cognitive Level: "Remember"

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3 Consider the grammar with the following translation rules and E as the start symbol. (1)
E → E1 # T { E.value = E1.value * T.value } | T{ E.value = T.value }

T → T1 & F { T.value = T1.value + F.value } | F{ T.value = F.value }

F → num { F.value = num.value }

Compute E.value for the root of the parse tree for the expression:

2 # 3 & 5 # 6 & 4.

140 150 160 260

Course Outcome: "CO3" / Cognitive Level: "Application"

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4 Error Recovery helps to (1)

Report Multiple errors	Rectify Multiple errors	Both report and rectify Multiple errors	None of this

Course Outcome: "CO2" / Cognitive Level: "Remember"

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5 Loops are the major targets for optimization since (1)

Loop may go to infinite execution	Loop body is repeated to several times	Condition check takes exceedingly large time	None of the other options

Course Outcome: "CO5" / Cognitive Level: "Understand"

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6 Which of the following statements about peephole optimization is False? (1)

It is applied to a small part of the code	It can be used to optimize intermediate code	To get the best out of this, it has to be applied repeatedly	It can be applied to the portion of the code that is not contiguous

Course Outcome: "CO5" / Cognitive Level: "Remember"

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7 A grammar with production rules { A → Ba | Cb, B → CA, C → c | ε } contains (1)

Left factor	Left recursion	Both left factor and left recursion	None of the other options

Course Outcome: "CO3" / Cognitive Level: "Understand"

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8 For the grammar rules {S → Aa | bB, A → c | ε}, (1)

FIRST(S) is

{b, c}	{a, b}	{a, b, c}	{a, b, c, ε}

Course Outcome: "CO3" / Cognitive Level: "Understand"

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9 Lex specification file sections are demarcated by (1)

%	{%	%}	%%

Course Outcome: "CO1" / Cognitive Level: "Remember"

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10 Output of lex program is available in a file named (1)

Lex.c	Lex.yy.c	Lex.l	Lex.yy.l

Course Outcome: "CO1" / Cognitive Level: "Remember"

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11 In shift-reduce parsing, handle is at (1)

Top of the stack	bottom of the stack	Anywhere in the stack	Nowhere in the stack

Course Outcome: "CO3" / Cognitive Level: "Remember"

12		

For the grammar
S → AB | C
A → bA | a
B → abbS | bS | ε
C → bC | ε

Follow(A) is

- a, $
- a, b, $
- a, b
- b, $

(1)

Course Outcome: "CO3" / Cognitive Level: "Understand"

13		

In Operator Precedence parsing handle is (1)

- Before <·
- After ·>
- Between <· and ·>
- None of the other options

Course Outcome: "CO2" / Cognitive Level: "Remember"

14		

Which of the following statements is true regarding LR parsers? (1)

- SLR and Canonical LR have the same number of states.
- LALR and Canonical LR have the same number of states.
- SLR and LALR have the same number of states.
- All three have the same number of states.

Course Outcome: "CO3" / Cognitive Level: "Understand"

15		

Amount of Look ahead in LALR parse is (1)

- 0
- 1
- 2
- 3

Course Outcome: "CO3" / Cognitive Level: "Remember"

16		

Which of the following conflicts is not possible in shift-reduce parsing (1)

- Reduce-reduce conflict
- Shift-reduce conflict
- Shift-shift conflict
- None of the other options

Course Outcome: "CO3" / Cognitive Level: "Remember"

17		

Which of the following is a lexical analysis tool (1)

- Lex
- Flex
- Jflex
- All of the other options

Course Outcome: "CO1" / Cognitive Level: "Remember"

18		

Intermediate code helps in (1)

- Program Analysis
- Code optimization
- Retargeting code
- Code check

Course Outcome: "CO4" / Cognitive Level: "Remember"

19		

A certain compiler corrects error like "fi" to "if" automatically. This is example of recovery in (1)

Panic mode   Delete character   Replace character   Transpose character

Course Outcome: "CO1" / Cognitive Level: "Understand"

20		

Which of the following is not a phase of a compiler? (1)

Lexical Analysis   Syntax Analysis   Code Optimization   Assembly Loading

Course Outcome: "CO1" / Cognitive Level: "Remember"

Section 2

Answer 5 out of 6 questions.

1		

Translate the following arithmetic expression into an appropriate intermediate representation: (5)

A = (B + C) * (D - E) + F / G

(a) Generate the corresponding **Three-Address Code (TAC)** and clearly describe each step involved in the translation.

(b) Represent the same expression using other intermediate representations, including **Quadruples**, **Triples**, and **Indirect Triples**.

Course Outcome: "CO4" / Cognitive Level: "Application"

2		

For the given regular expression: (5)

((a|b)a)*b

Perform:

1) Syntax tree construction and position labeling.
2) nullable(), firstpos(), lastpos(), and followpos() calculation.
3) Construct the NFA step by step.

Course Outcome: "CO2" / Cognitive Level: "Application"

3		

Enlist and discuss the challenges encountered during the Syntax Analysis phase of the compiler. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

4		

Compute the operator precedence relation graph and precedence relation table for given grammar. (5)

G → E
E → E+T | E-T
T → T*F | T/F
F → id

Course Outcome: "CO3" / Cognitive Level: "Analysis"

5		

Consider the following grammar: (5)
E-> E+T | T
T-> TF | F
F-> F* | a | b

Construct the CLR parsing table and also parse the input "a*b+a". Identify is there any conflict.

Course Outcome: "CO3" / Cognitive Level: "Understand"

6		

Describe how the symbol table is used during first three phases of compilation with suitable example . (5)

Course Outcome: "CO2" / Cognitive Level: "Understand"

Section 3

Answer 5 out of 6 questions.

1		

Draw the syntax tree and DAG for the following expression: (5)

(a*b)+(c-d)*(a*b)+b

Course Outcome: "CO4" / Cognitive Level: "Application"

2		

Design an SDD for converting infix expressions to postfix expressions. (5)
Use a grammar such as:

E → E + T | T
T → T * F | F
F → (E) | id

Show the output of your translation for the expression (a + b) * c.

Course Outcome: "CO3" / Cognitive Level: "Application"

3		

The following code segment performs computations on multiple arrays using arithmetic operations within iterative structures: (5)

```
void compute(int a[], int b[], int c[], int d[], int n, int m) {
int x, y, z, sum = 0;

for (int i = 0; i < n; i++) {
x = a[i] * m;
y = b[i] + m;
z = x + y;
c[i] = z;
d[i] = c[i] * 2;

// dead code (never used)
int temp = a[i] * 10;
}

for (int i = 0; i < n; i++) {
sum += d[i];
```

```
}

printf("Sum: %dn", sum);
}
```

1) Carefully examine the above code and identify any sections that could be improved for better performance and efficiency.

2) Justify your improvements by explaining what parts of the code are redundant, repeated unnecessarily, or can be simplified.

Course Outcome: "CO5" / Cognitive Level: "Create"

4		

Show that the grammar, where e is epsilon (5)
S -> AaAb | BbBa
A -> e
B -> e

is LL(1) but not SLR(1).

Course Outcome: "CO3" / Cognitive Level: "Application"

5		

Discuss any two garbage collection techniques with advantages and disadvantages. (5)

Course Outcome: "CO5" / Cognitive Level: "Understand"

6		

Explain the concepts of parse tree, annotated parse tree, and syntax tree using below expersstion. How do they differ in structure and purpose during the syntax analysis phase of compilation? (5)

5*10+6-12/6+9

Course Outcome: "CO2" / Cognitive Level: "Understand"

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Event Name	Subject
University Exam April-May 2025	CSE313:DESIGN OF LANGUAGE PROCESSOR

#	Course Outcome	Description	Marks
1	CO1	Understand the fundamentals of the compiler identify the relationships among different phases of the compiler and use the knowledge of the Lex tool	7
2	CO2	Describe theRole of Parser and the various error recovery strategies	17
3	CO3	Develop the parsers and experiment with the knowledge of different parser designs.	33
4	CO4	Develop a semantic analysis scheme to generate intermediate code	11
5	CO5	Summarize various optimization techniques used for dataflow analysis and generate machine code from the source code	12
		Total	80

* NA - CO Not Selected

Event Name	Subject
University Exam April-May 2025	CSE313:DESIGN OF LANGUAGE PROCESSOR

#	Level	No of Questions	Marks (%)

1	Remember	12	12(15%)
2	Understand	11	27(33.75%)
3	Application	7	31(38.75%)
4	Analysis	1	5(6.25%)
5	Create	1	5(6.25%)
6	Evaluate	0	0(0%)
	Total	32	80 (100%)